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Chapter 01 - Number Systems.

* Most commonly used number system is Decimal number system. This system as its name indicates, is based on the number 10

The number 10 is called the basis or radix of the system

To indicate the number in the system we used ~~dee~~ the symbols : 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

This symbols are called digit

When using numbers in some other ~~ways~~ base we will used $(x)_\beta$ to indicate that the number x is to be interpreted in the base β number system

The binary Number system

This system is based on the number 02
The digits are 0, 1 $\therefore \beta = 2$

Examples: $(1101)_2$, $(1111)_2$, $(11011.0\overset{2^{-1}}{1}11)_2$

The octal Number System

This system is based on the number 08
The digits are 0, 1, 2, 3, 4, 5, 6, 7 $\therefore \beta = 8$

Examples $(503)_8$ $(1234)_8$, $(5413.\overset{8^{-1}}{7}06)_8$

The Hexa Decimal Number System

This system is based on the number $(16 = \beta)$
~~The~~ ~~d~~ Digits are 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F

$9/2 = 2 + \frac{1}{2}$

The reason for choosing letters instead of numerical symbols to keep all the symbols combination of in single characteristics.

ex:- $(56C.F)_{16}$, $(1A6F)_{16}$

Let β be a positive integer. Then β different symbols (digits) are used to write a number

Convert in decimal to other ~~number system~~ bases

(i) Decimal to binary

Let x be a positive decimal number (integer)

~~Let $x = (a_n, a_{n-1}, a_{n-2} \ldots$~~

$(n+1)$ symbols

Let $x = (a_n \; a_{n-1} \; a_{n-2} \ldots \ldots a_1 \; a_0)_2$

$\Rightarrow x = a_n \times 2^n + a_{n-1} \times 2^{n-1} + a_{n-2} \times 2^{n-2} + \ldots \ldots + a_1 \times 2^1 + a_0 \times 2^0$

$\frac{x}{2} = x_0$

$$\frac{x_0}{2} = \underbrace{a_n \times 2^{n-1} + a_{n-1} \times 2^{n-2} + a_{n-2} \times 2^{n-3} + \ldots + a_1}_{x_1 = \text{Quotient}} + \frac{a_0}{2}$$

$$\frac{x_1}{2} = \underbrace{a_n \times 2^{n-2} + a_{n-1} \times 2^{n-3} + a_{n-2} \times 2^{n-4} + \ldots + 2^0 a_2}_{x_2} + \frac{a_1}{2}$$

$$\frac{x_2}{2} = \underbrace{a_n \times 2^{n-3} + a_{n-1} \times 2^{n-4} + a_{n-2} \times 2^{n-5} + \ldots + 2 a_3}_{a_3} + \frac{a_2}{2}$$

$2+\frac{1}{2}$

$$\frac{x_{n-1}}{2} = \frac{a_n}{x_n} + \frac{a_{n-1}}{2}$$

$$\frac{x_n}{2} = \frac{a_n}{2} + 0$$

+ Now it is clear that a_i is the remainder when x_i is divided by 2 for

$a_i = 1, 2, \ldots, n$

This procedure is performed until we get the zero quotient

This procedure may be extended any other ~~positive~~ decimal number system other ~~equivalents~~. equivalents. such as octal, hexadecimal etc.

Example :

What is binary equivalent of decimal 41?

$$\begin{array}{r|l}
2 & 41 \\
2 & 20 - 1 \\
2 & 10 - 0 \\
2 & 5 - 0 \\
2 & 2 - 1 \\
 & 1 - 0
\end{array}$$

$(101001)_2$

~~45~~ ~~41~~

Octal equivalent of Decimal 41 ?

$$\begin{array}{r|l}
8 & 41 \\
8 & 5 - 1 \\
 & 0 - 5
\end{array}$$

$(51)_8$

0.01 2^{-1} 2^{-2}

$0.5 \times 5 +$ 7×0.25 2^{-1} 2^{-2}

0.1 $0.(5)$

Hexa decimal equivalent of 41?

$$\begin{array}{r|l} 16 & 41 \\ 16 & 2 - 9 \\ & 0 - 2 \end{array}$$

$(29)_{16}$

How to convert a positive decimal fraction ~~to its~~ $x (<1)$ to its binary equivalent or its other equivalent

Let $x = (.a_1 a_2 a_3 \dots a_n)_2$

$$= \underbrace{a_1 \times 2^{-1} + a_2 \times 2^{-2} + a_3 \times 2^{-3} \dots + a_n \times 2^{-n}}_{x_1}$$

$$2x_1 = a_1 \times 2^0 + \underbrace{a_2 \times 2^{-1} + a_3 \times 2^{-2} \dots + a_n \times 2^{-n+1}}_{x_2}$$

$$2x_2 = a_2 + \underbrace{a_3 \times 2^{-1} + a_4 \times 2^{-2} + \dots + a_n \times 2^{-n+2}}_{x_3}$$

$x = (.a_1 a_2 a_3 \dots a_n)_\beta$

$$= a_1 \times \beta^{-1} + a_2 \times \beta^{-2} + \dots + a_n \times \beta^{-n}$$

$$\beta x = a_1 + \underbrace{a_2 \times \beta^{-1} + \dots + a_n \times \beta^{-n+1}}_{x_1}$$

$\uparrow$ Integer part of βx.

$$\beta x_1 = a_2 + a_3 \times \beta^{-1} + \dots + a_n \times \beta^{-n+2}$$

$\uparrow$ Integer part of βx_1

Wise

0.1 ×2 = 0.2
0.2 ×2 = 0.0

0.1
0.75×2 = 1.50 0.75
0.50×2 = 1.00
2.75

What is the binary representation of decimal $x = 0.1$

Converting	Results
0.1	0
0.1×2 = 0.2 = 0+0.2	0.0
0.2 ×2 = 0.4 = 0+0.4	0.00
0.4×2 = 0.8 = 0+0.8	0.000
2×0.8 = 1.6 = 1+0.6	0.0001
2×0.6 = 1.2 = 1+0.2	0.00011
2×0.2 = 0.4 = 0+0.4	0.000110
2×0.4 = 0.8 = 0+0.8	0.0001100
2×0.8 = 1.6 = 0+0.6	0.00011001
2×0.6 = 1.2 = 1+0.2	0.000 110011
2×0.2 = 0.4 = 0+0.4	0.000 1100110

Wi

Date No.

Floating Point number systems.

* Although practical problems deal with real valued quantities & theroms upon which numerical methods are based are written in terms of real valued functions, computers like measurement devices have no concept of the real numbers. Instead, computers represent real numbers in what is known as a floating point number systems.

Definition

A floating point number system $F(\beta, k, m, M)$ is a subset of real number system characterized by the parameters.

β - the Base
k - the number of digits in the base β expansion
m - the minimum exponent
M - the maximum exponent
e : exponent

elements of $F(\beta, k, m, M)$ are those real numbers that can be expressed exactly as

$$f = \pm (0.\, d_1\, d_2\, d_3 \ldots d_k)_\beta \times \beta^e$$

where $0 \leq d_k \leq \beta - 1$ and $m \leq e \leq M$

Also, note that when $f \neq 0$, $d_1 \neq 0$, but $f = 0$, $d = 0$

$$(0.\, d_1 d_2 d_3 \ldots d_k)_\beta = \frac{d_1}{\beta} + \frac{d_2}{\beta^2} + \frac{d_3}{\beta^3} \ldots + \frac{d_k}{\beta^k}$$

is called the mantissa of the floating point number.

$10-(0-9)$

(x, y)

The number β is called radix and d_1 is called radix point

example

consider the system $f(\overset{\beta}{10}, \overset{k}{2}, \overset{m}{0}, \overset{M}{2})$

Here $\beta = 10, k=2, m=0$, and $M=2$.

$0 \le e \le 2$, $e = 0, 1, 2$

Also so, Digits in base β system

: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

$f = \pm(0 . d_1 d_2) \times 10^e$, if $f \neq 0$

d_1	d_2
1	0
2	1
3	2
4	3
5	4
6	5
7	6
8	7
9	8
	9

When $e = 0$

$\pm(0 . d_1 d_2) \times 10^0 = \pm(0.d_1 d_2)$

: $\pm(0.10)$, $\pm(0.12)$, $\pm(0.1$
$\pm(0.13)$, $\pm(0.14)$, $\pm(0.1$
$\pm(0.16)$, $\pm(0.17)$, $\pm(0.1$
$\pm(0.19)$, $\pm(0.21)$, $\pm(0.2$
$\ldots \pm(0.29)$

$\pm(0.90)$ $\pm(0.91)$; $\ldots \pm(0$

The number of elements $= 2 \times 90 = 180$

when $e=1$

$\pm(0.d_1d_2)\times 10^1 = \pm(d_1.d_2)$

$\therefore \pm 1.0, \pm 1.1, \pm 1.2 \ldots \pm 1.9$

$\vdots$

$\pm 9.0, \pm 9.1 \ldots \pm 9.9$

The number of elements $= 2\times 90 = 180$

when $e=2$

$\pm(0.d_1d_2)\times 10^2 = \pm(d_1d_2)$

$\therefore \pm 10, \pm 11, \pm 12 \ldots \pm 19$

$\vdots$

$\pm 90, \pm 91, \pm 92 \ldots \pm 99$

The number of elements $= 2\times 90 = 180$

Also note that $0 \in F(10,2,0,2)$

$\therefore$ The total number of element $= 3\times 180 + 1 = 541$

Round off errors

Suppose that the floating point number system $F(10, 2, 0, 2)$ is used in a hypothetical computer

$$\boxed{\begin{array}{ll} \beta = 10 & M = 2 \\ K = 2 & \\ m = 0 & \end{array}}$$

* A given calculations requires the value
$\sqrt{7.1} \approx 2.66458$

* Since the number has more than two digits it is not an element in $F(10, 2, 0, 2)$

* ~~How~~ Now how can represent the number in this number system computer
$2.66458 \notin F(10, 20, 2)$
This may be performed using either of following

1) Dropping all digits after the second digits
* take the approximation ~~square~~ to
$\sqrt{7.1} \approx 2.6$ this approach is called chopping the numbers

2) Rounding the numbers to two digits of accuracy, Take the approximation $\sqrt{7.1} \approx 2.7$ this approach is called rounding the number

3) The general, let y be a real number whose expansion is given by ~~$y = \pm(0.d_1d_2 \dots d_k$~~

$$y = (0.d_1d_2 \dots \dots d_k\ d_{k+1}\ d_{k+2})_\beta \times \beta^e$$ wr

and $m \le e \le M$.

Let $fl(y)$ denote the floating point equivalent of y. This is $fl(y) \in F(\beta, k, m, M)$.

When the number is chopped

$$fl_{cop}(y) = \pm (0 . d_1 d_2 \ldots d_k)_\beta \times \beta^e$$

When the number is rounded.

$$\cancel{fl_{round}(y) = \pm (0 . d_1 d_2 d_3 \ldots d_k}$$

$$fl_{round}(y) \begin{cases} \pm (0 . d_1 d_2 d_3 \ldots d_k)_\beta \times \beta^e & \text{if } d_{k+1} < \beta/2 \\ \pm \left[(0 . d_1 d_2 d_3 \ldots d_k)_\beta + \beta^{-k} \right] \times \beta^e & \text{if } d_{k+1} \geq \beta/2 \end{cases}$$

Here $\beta^{-k} = (0.0020 \ldots 0)_\beta$

Definition

The error introduced by converting earlier a real number to its floating point equivalent is called Round off error

Definition

Let p^* denoted any approximation to the number p. The absolute error in p^* is given by $|p-p^*|$

The relative error in p^* is given by

$$\left|\frac{p-p^*}{p}\right|$$

Floating point Arithmetic

Let x and y ~~we~~ be two real numbers. Let $\oplus, \ominus, \otimes$ and $\oslash$ denoted the corresponding floating point equivalent operations of arithmetic operations $+, -, \times$ and $\div$

Real Number system

$+$	$\times$	x
$-$	$\div$	y

Floating Number system

$\oplus$	$\otimes$
$\ominus$	$\oslash$
$x \oplus y$	

Now floating point operations are defined

$$x \oplus y := fl[fl(x) + fl(y)]$$

$$x \ominus y := fl[fl(x) - fl(y)]$$

$$x \otimes y := fl[fl(x) \times fl(y)]$$

$$x \oslash y : fl(fl(x) \div fl(y))$$

Ex. Using four decimal digits rounding arithmetic find $5/3 \otimes \sqrt{3}$

$5/3 = 1.66667$

$\sqrt{3} = 1.73205$

$5/3 \otimes \sqrt{3} = 1.667 \otimes 1.732$

$= \cancel{fl_{round}[fl_{round}(1.667) \times fl_{round}(1.732)]}$

$= fl_{round}[fl_{round}(1.666|67) \times fl_{round}(1.732|05)]$

$= fl_{round}(1.667 \times 1.732)$

$= fl_{round}(2.887|244)$

$= 2.887$